

A Lightweight Machine Learning Approach for Detecting AI-Generated Phishing Emails in SIEM Environments

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Abstract

AI generation for phishing can be very dangerous because it is more convincing and blends in better than what traditional security filters can handle. Even with the help of existing tools, some can be a heavy load and slow down work procedures. This research aims to provide a simpler and more lightweight solution that is strong enough to detect possible threats without relying on heavy, complex neural networks. Beyond detection, we also implement a Security Information and Event Management (SIEM) system to log activity and alert Security Operations Centers (SOCs). The goal is to make the workflow in enterprise environments more productive and faster.

Keywords— AI-Generated Phishing, Machine Learning, SIEM Integration, Logistic Regression, Cybersecurity, Security Operations Center (SOC), Generative AI.

1 Introduction

Email continues to serve as one of the primary communication channels for organizations or company, but it also remains a major entry point for cyberattacks, particularly phishing. Traditional anti-phishing defenses depend on rule-based filtering, sender reputation checks, and signature-matching systems. However, these techniques are becoming increasingly ineffective due to the rise of AI-generated phishing emails, which appear linguistically natural, diverse, and highly convincing.

Recent studies demonstrate that generative AI enables attackers to craft large volumes of unique phishing messages that evade pattern-based detection systems. As shown in the work of Eze and Shamir (2024), AI-generated phishing emails differ from human-generated phishing emails in vocabulary, structure, sentiment, and stylistic patterns, making them detectable by machine learning but difficult for rule-based filters to identify reliably.

To address these challenges, this research pro-

poses a lightweight machine learning–assisted phishing detection framework designed to identify AI-generated phishing emails and demonstrate how detection outputs can be integrated into a Security Information and Event Management (SIEM) environment. The proposed approach emphasizes computational efficiency and practical feasibility, enabling effective analysis of suspicious emails without introducing significant processing overhead. By producing structured detection outputs suitable for SIEM ingestion, the framework aims to support SOC workflows and improve centralized visibility for incident analysis and response.

2 Background or literature review

Phishing has historically relied on social engineering to deceive users into providing credentials, transferring funds, or performing unauthorized actions. Before generative AI, most phishing attacks reused identical or near-identical email templates. These could be detected through signature matching, metadata analysis, URL filtering, or identifying duplicates across recipients.

2.1 Evolution of Phishing Detection

Phishing attacks, first emerging in the mid-1990s with simple deceptive emails targeting AOL users, have evolved dramatically in sophistication and scale. Early attacks relied on crude, mass distributed messages with obvious grammatical errors and generic content. Over time, they progressed to spear-phishing highly targeted campaigns leveraging personal information and more recently, to AI-assisted attacks that generate personalized, contextually coherent messages at scale. This escalation has driven parallel advancements in detection methodologies, progressing through distinct phases to counter increasingly evasive threats.

Traditional approaches include:

- Header analysis (sender, IP, domain mismatch)
- URL blacklist/whitelist methods

- Statistical text analysis
- Machine learning using classical algorithms such as SVM, Random Forest, Naïve Bayes
- Deep learning approaches including CNN, RNN, and LSTM frameworks

While these methods are effective against conventional phishing emails, they are less capable of detecting unique, AI-generated messages that do not match known signatures.

2.1.1 AI-Generated Phishing Emails

Eze and Shamir (2024) demonstrated that generative AI platforms can automatically generate numerous phishing messages that are grammatically correct and highly contextually coherent. Their study produced a corpus of 865 AI-generated phishing emails and used multiple machine learning models—including MALLET, UDAT, and LSTM networks—to classify these emails. Results showed very high accuracy (up to 100% in some models) due to measurable differences between AI-generated and human-crafted phishing emails, particularly in vocabulary diversity, word length, pronoun frequency, and sentiment distribution.

2.1.2 Gap in Existing Research

Despite these findings and ability to detect AI-generated texts, there remains a considerable gap in actual enterprise security efforts. For example, there exists an AI Module which can detect AI-generated phishing, but it's an isolated detection effort with many false positives not yet integrated into a cohesive security ecosystem. One of the most significant shortcomings in the research is the emphasis on offline text classification which does not take into account the text parsing limitations for processing live email traffic in real-time to avoid slowing down production. In addition, there is no clear automated integration with SIEM that would provide the SOC with centralized visibility and correlation. Finally, the research does not provide a structured SOC workflow that would allow a SOC analyst to engage with AI-related alerts. This research proposal seeks to bridge

this gap by proposing a Security Operation pipeline that integrates AI detection with SIEM logging and SOC incident response.

3 Problem Statement

The fact that the use of emails as the medium for carrying out communication has become widespread means that emails are a fertile ground for phishing attacks. However, recently, the nature of phishing attacks has been made more sophisticated by the use of generative AI. This technology can enable the attackers to produce a tremendous amount of phishing emails that are fluent in language, context-conscious, and tailored, hence making it difficult for the traditional rule-based solutions for blocking phishing emails that depend on static signatures, known indicators, or repetitive templates to detect the phishing emails before they reach their intended targets.

Though machine-learning approaches have proved their merits regarding detecting phishing content, their implementation in an organizational setup is still a challenge. This is because a complex machine-learning algorithm may cause processing delays that can affect legitimate emailing processes, whereas a simple algorithm may not provide adequate detection of AI-phishing content. Furthermore, most of the existing approaches regarding phishing detection are standalone systems and do not provide effective integration with Security Information and Event Management (SIEM) systems. This is a hindrance to gaining a centralized Overview regarding this process and reacting accordingly from a Security Operations Center (SOC).

In addition, even if phishing emails could be effectively identified by machine learning algorithms, a structured process may not exist within current SOC practices for handling such detections. With this, merely detecting phishing emails may not effectively address the risk that AI-generated phishing attacks present, even if effectively identified, because alerting and prioritization systems and decision-making assistance may not be provided. Thus, a lightweight phishing detection method that leverages machine learning and can effectively distinguish AI-generated

phishing emails while operating efficiently and integrating with SIEM systems for effective SOC monitoring is also required.

4 Research Objectives

This research aims to investigate the feasibility of a lightweight machine learning-based email detection approach for identifying AI-generated phishing emails and demonstrating its integration within a SIEM-oriented monitoring context. The specific objectives of this research are as follows:

1. To implement a text-based machine learning classifier capable of distinguishing AI-generated phishing emails from legitimate emails.
2. To apply minimal preprocessing and feature extraction techniques suitable for fast deployment in SOC environments.
3. To generate phishing risk scores that can be consumed by a SIEM system for centralized logging and alerting.
4. To evaluate the effectiveness of the proposed approach using standard classification metrics such as accuracy, precision, recall, and F1-score.
5. To demonstrate the feasibility of integrating machine learning-based phishing detection into SIEM workflows without introducing significant operational overhead.

5 Methodology

This research work makes use of an experimental approach to evaluate the practicality of using a lightweight machine learning approach in detecting phishing emails generated by means of AI and to validate its integration with respect to monitoring in a SIEM-oriented setup. The proposed approach to this problem has been divided into a series of stages, which include dataset development, text pre-processing, feature extraction, machine learning classification, development of SIEM-compatible outputs, and evaluation. This will ensure that the proposed approach

is computationally viable, reproducible, and SOC-ready without involving complexities related to full-scale product implementation.

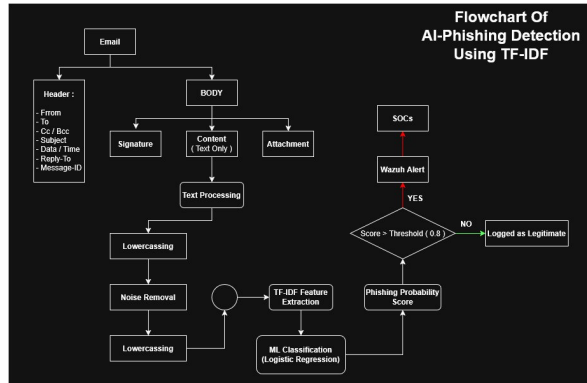


Figure 1: System Flowchart of the AI-Phishing Detection Pipeline

5.1 Dataset Description

The research uses publicly available email dataset examples, which include phishing as well as legitimate emails (also referred to as non-phishing emails), for the supervised machine learning analysis. The presence of both phishing and non-phishing emails allows it to distinguish malicious from normal email interactions.

The phishing emails, examples of which can be assumed to cover modern phishing attacks that may be produced or enhanced using generative AI techniques, may be well-crafted, contextually informed, and linguistically natural, just like modern AI-generated phishing attacks, as described in some of the most recent research works. Authentic emails may also be incorporated, examples of which may cover common organizational communications, ensuring that the classification boundary is not overly restrictive.

For each email occurrence, the subject and body of the email are merged together as a single textual feature. In this approach, the model is able to assess the entire information in an email without considering the subject and body separately. The produced

text data can be used as inputs for the process of feature extraction.

The dataset is only used for academic and research purposes. None of the identifiable information found in the original dataset is preserved or examined. The resulting dataset is divided into two parts, namely training and test, to assess the capability of the proposed machine learning approach to identify phishing emails generated by AI.

Phishing emails were obtained from the AI Phishing Email Dataset (https://people.cs.ksu.edu/~lshamir/data/ai_phishing/), while legitimate emails were sourced from the Enron Email Dataset (<https://www.cs.cmu.edu/~enron/>).

5.2 Text preprocessing

Text processing is carried out to make the raw text of the emails suitable for analysis using machine learning techniques. The aim of this process is to dampen the noise present in the text data and retain useful linguistic patterns relevant to the identification of phishing emails created through AI techniques. A light processing method is used for efficiency and SOCs.

First, all text within the email is normalized to lowercase to guarantee consistency and omit identical words with varying cases from being recognized as separate key phrases. Additionally, all punctuation and special characters are removed because they do not play a key role in recognizing phishing key phrases during text classification problems. Stop words like “the,” “and,” and “is” are also eliminated for dimensionality reduction purposes and to focus on more significant key phrases.

There is no advanced linguistic processing, such as stemming or lemmatization. This is because over-processing might result in masking the stylistic and lexical markers in AI-produced phishing emails. The preprocessed texts for each email feature set then move to the feature extraction phase for their numeric characterization.

5.3 Feature Extraction

The final step of preprocessing involves transforming the cleaned text of the emails into a form that can

be processed by the machine learning algorithm. For this purpose, in this paper, Term Frequency-Inverse Document Frequency (TF-IDF), which is simple and suitable for lightweight machine learning algorithms, has been used for feature extraction.

TF-IDF calculates the representation for each email using a vector that indicates the importance of the words used in the corpus. Term Frequency (TF) calculates the frequency of each word used in each email, while the Inverse Document Frequency (IDF) reduces the weight of terms that are used frequently across the entire corpus. The two stages help to ensure that the model gives much importance to terms that are used to create phishing pages.

TF-IDF makes it easier to identify the subtle variation in word usage that is characteristic in phishing emails generated by AI systems, such as repeated use of urgency-related words, persuasive structuring of sentences, and/or terminology particular to the context among others. In light of more complex representations of extracted features, TF-IDF is characterized by simpler computations and better interpretability.

The TF-IDF vectors obtained are used as inputs for a machine learning model for classification in the next stage.

5.4 Machine learning

A supervised machine learning approach is used for the classification of the phishing and legitimate emails. In this study, the proposed analysis conducts the classification using Logistic Regression because this approach is computationally inexpensive and easy to understand, and it is suited for small-scale network monitoring tasks.

The logistic regression model attempts to establish a relationship between the TF-IDF vectors and their corresponding email labels by weighing the textual features that are characteristic of phishing emails. The patterns derived from this training allow a probability result that determines the likelihood of a particular email being a phishing attempt.

The probabilistic result, rather than a binary outcome, supports threshold alerting, which when combined with the use of a SIEM system, makes this

possible. This setup allows Security Operations Center analysts to tune the sensitivity levels per the risk tolerance levels that the organization prefers to assume. Compared to other machine learning approaches, such as deep learning, Logistic Regression trains and conducts inference faster and still manages to effectively detect phishing messages created via AI.

5.5 SIEM Integration

To demonstrate the integration of machine learning-assisted phishing detection results into security monitoring tasks, outputs of the classifier are formatted in a form suitable for consumption and processing by a Security Information and Event Management System. Contrary to incorporating a machine learning module into a Security Information and Event Management System, an approach based on the indicated methodology considers a machine learning module as an external component.

The machine learning model produces a probability-based phishing risk score and a corresponding class for every email that is analyzed. The result is represented as a log message that contains relevant metadata such as the email ID, time of detection, predicted class, and the associated risk score. The log message format is what SIEMs typically consume.

In the context of the SIEM solution, the alerting function based on thresholds is used to handle phishing risk scores. Messages that have scores above a certain threshold are considered to be high-risk phishing incidents, thus allowing SOC's to prioritize and address them effectively. By integrating in such a manner, visibility is improved without compromising the simplicity or complexity inherent in full system deployments.

5.6 Evaluation Metrics

The effectiveness of the suggested machine learning-based phishing detection system is analyzed by using traditional measurement of classification commonly utilized in the security domain and machine learning. Such metrics provide an understanding of the

ability of the model to recognize AI-generated phishing emails with the highest degree of accuracy and at the same time, reduce false alarms.

Accuracy measures the total correctness of the results of the classification. Precision measures the percentage of cases labeled phishing that are actually phishing and thus reduces any effects of false-positive warnings which can be a burden on SOC analysts. Recall is a ratio of the actual number of phishing mail that is recognized correctly, showing the effectiveness of the model in notifying the possibility of a threat. The F1 score is also given, which gives a harmonic mean that balances between precision and recall.

Taken together, these assessment indicators allow conducting a thorough evaluation of the effectiveness and feasibility of the suggested method of identifying phishing emails generated by AI in a SIEM-based monitoring setup.

6 Result

The intended outcomes of the current research are that the suggested lightweight machine learning system is likely to be efficient in detecting AI-generated phishing email and be operable in the environment of the Security Operations Center. It is believed that the Logistic Regression classifier, which is being trained on the basis of TF-IDF features of the slightly preprocessed text in emails, will demonstrate good classification rates, high accuracy, precision, recall, and F1-score value similar to others reported in recent AI-phishing detection research. The resulting probabilistic risk scores of the model are anticipated to give significant distinction between high-risk phishing emails and genuine messages so that alerting within a SIEM system can be configured with flexibilities in the setting of threshold parameters. Moreover, the minimal computational cost of the preprocessing, feature extraction, and classification methods chosen is likely to enable the analysis of emails in near real time without adding any similar delays to the usual functions of emails. The framework will improve centralized visibility and enable events correlation as well as assist SOC analysts to prior-

itize phishing-related incidents by formatting detection outputs as structured SIEM-compatible logs. In general, the outcomes will prove the hypothesis that it is possible to effectively detect AI-written phishing emails with the help of lightweight and interpretable machine learning methods, which can be easily incorporated into the existing enterprise security monitoring procedures.

7 Conclusion

The use of generative AI has made the conventional rule-based and signature-based email security control mechanisms ineffective due to the increased linguistic sophistication, situational awareness, and diversity of phishing. According to the description in this study, these aspects allow the phishing messages produced by AI to get beyond standard detection tools and increase the threat to the security of an organization.

This study was effective in revealing that a simplified machine learning model can be utilized to identify AI-generated phishing emails with the output that can be utilized in a SIEM monitoring environment. Small amount of text pre-processing combined with TF-IDF vectorization and a Logistic Regression classifier adopted by the classifier fitted well with low-computational needs but still offered satisfactory phishing classification. It is a design that can be used in a practical setting in a Security Operations Center (SOC) that focuses on the importance of performance, scalability, and responses in a timely manner.

The operational focus of this research, as opposed to an independent detection characteristic, is one of the primary contributions of the research. Probability-based phishing risk scores and SIEM-friendly logs can be generated and centralized visibility, thresholded alerts, and prioritization may be done within the SOC. This offers a pathway of bridging the gap in the problem statement between real world enterprise security operations and machine learning-based phishing detection.

Overall, this study shows that a lightweight machine learning system can be useful in detecting attributes of AI-generated phishing emails without being complicated or data-intensive. In this way, the

suggested framework provides a reasonable equilibrium between detection efficiency and operational efficiency which satisfy the requirements of Security Operation Center (SOC). The framework could be improved in the future by adding more contextual features to the approach, performance measurement within a live-email setting, or better thresholding methods to minimize false positives and enhance SOC responsiveness.

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